

Vehicle Type Classification

Dataset And Research Question

We will use Stanford Cars Dataset which contains 16,185 images of 196 classes of cars. The main research questions are: 1. How accurately can traditional ML methods classify these images? 2. What performance gains do deep learning approaches provide compared to traditional methods (SVM vs. CNN) for this case?

Methods And Performance Evaluation

One approach is to use Support Vector Machine with a Histogram of Oriented Gradients features. HOG is used to capture edge orientations and shape information, which we hypothesize will work well for car body shapes but may perform poorly for finer classification such as badges on grills.

The second approach is to use Convolutional Neural Networks. The CNN will learn hierarchical features automatically through backpropagation. Both methods will use identical train/validation/test splits (70/15/15 of the training set).

Both methods will be evaluated using Top-1 and Top-5 accuracy to measure overall classification performance. By manually examining the top 20 misclassifications from each method we can also analyze the errors to understand common failure modes.

Timeline/Team Collaboration

Weeks 1-2: Researching basics + setting up environment for developing the model (including importing and working with Stanford Cars Dataset), extract HOG features, train SVM, get first results

Week 3-4: Analyze errors, tune SVM parameters, begin research into CNNs/setting up Pytorch or TensorFlow

Weeks 5-7: Build simple CNN architecture, get it training. Debug, tune hyperparameters. Train final model, fix any issues(overfitting)

Weeks 8-9: Analyze and compare models, analyze errors, researching/understanding why performance differs. Can also be used as buffer in case CNN needs more work

Week 10: Write-up

References

Krause, J., Stark, M., Deng, J., & Fei-Fei, L. (2013). 3D Object Representations for Fine-Grained Categorization. *IEEE Workshop on 3D Representation and Recognition*.

Dalal, N., & Triggs, B. (2005). Histograms of Oriented Gradients for Human Detection. *CVPR*.